Evidence for age-related improvements in language prediction

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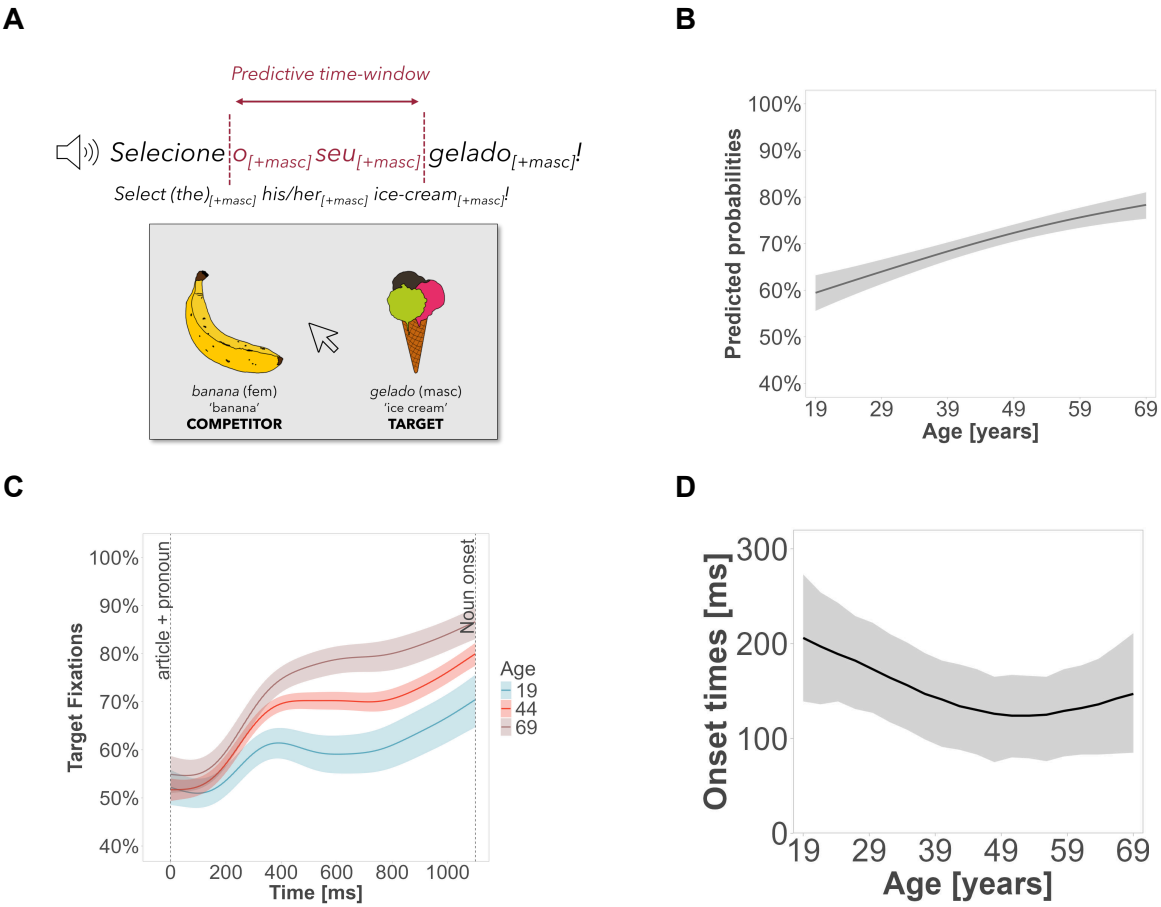
Comprehenders use agreement relationships to predict upcoming sentence constituents [1,2,3]. Previous work suggests that predictive processing is modulated by non-linguistic and linguistic abilities. For instance, comprehenders with higher working memory capacity or higher language experience show stronger or faster predictive effects [2,3]. However, most evidence comes from studies with young adults leaving a critical gap in understanding how predictive processing changes over the adult lifespan. Moreover, aging brings changes in individual factors that are likely to modulate predictive processing. This includes a decline in working memory and processing speed [4], but a preservation of some aspects of syntactic processing [5,6], likely due to compensatory mechanisms that result from lifelong language experience [7,8]. In our study, we used a continuous-age design with a well-distributed sample of participants to examine the use of morphosyntactic relationships for language prediction across the lifespan.

Method. We tested 135 European Portuguese native speakers (age range: 19–69 years) in a visual world eye-tracking experiment. In each trial, participants saw two objects while hearing an auditory instruction that included a determiner and a possessive pronoun (e.g., *o seu* '(the)[+masc] his/her[+masc]'), both expressing gender information that allowed predicting the target noun (e.g., *gelado* 'ice cream'[+masc]) (Fig. 1A). The eye-tracking data was analysed with a novel GAMM-based method [9] that allowed estimating the effects of age on three complementary measures: (i) overall proportion of looks to the target object during the predictive time-window; (ii) timecourse of fixations over time; and (iii) temporal onset of predictive effects. Participants also completed working memory and processing speed tasks.

Results and discussion. Our results revealed (i) an increase of predictive looks to the target object with increasing age (Fig. 1B), (ii) a significant age effect on the timecourse of fixations (Fig. 1C), and (iii) faster prediction onsets for older than younger adults, at least until the mid-50s (Fig. 1D). Sensitivity analyses showed that the effects were robust, being observed both when controlling for working memory and processing speed and when age was unadjusted for any covariates. Thus, the observed age-related improvements occurred even alongside declines in these non-linguistic cognitive abilities. Overall, our findings suggest that increasing age can enhance the use of morphosyntactic cues to predict upcoming linguistic information. We propose that lifelong language experience strengthens the links between the noun's lemma

and gender nodes, enabling gender information to be activated more quickly during prediction. These results support the view that some sentence processing mechanisms can not only be well-preserved, but might even improve at older ages [5,7,8].

Figure 1: (A) Sample experimental trial. The images depict objects that differ in grammatical gender (e.g., flor 'flower'[+fem] vs. prato 'plate'[+masc]). Participants were introduced to the gender-neutral character *Glipi*, and asked to find their belongings by selecting the target object mentioned in the auditory instruction. Age effect on: (B) predictive looks to the target object; (C) timecourse of fixations across all ages and for the minimum, mean and maximum ages in the samples; (D) predictive onset times. The ribbons show 95% confidence intervals.



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